ORBITAL ACTIVE FIRE HARMONIZATION SYSTEM • ENGINE V2.1

Triton Wildfire

Bridging 24 years (2002–2026) of 1 km MODIS and 375 m VIIRS satellite thermal anomaly observations via spherical geodesic co-location, off-nadir bowtie scan suppression, and pure-physics radiometric calibration.

DATA STREAM: FIRMS LIVE NRT REST API

SENSORS: TERRA / AQUA / S-NPP / NOAA-20 / NOAA-21

SYSTEM TECHNICAL SPECIFICATION

Reference Baseline	MODIS C6.1 (3.96 µm, 1.0 km)
High-Res Stream	VIIRS I4 (3.74 µm, 375 m)
Geodesic Kernel	r ≤ 1.11 km (Haversine)
Empirical Validation	5 Regions / 24 Fire Fronts (KZ)
Source Repository	github.com/pulik181/triton-wildfire

SPATIAL FOOTPRINT RATIO

7.11 : 1

1 km² MODIS nadir cell unified with 375 m VIIRS sub-pixels (100 ha vs 14.06 ha)

RADIOMETRIC AGREEMENT

99.65%

Pooled Fire Radiative Power (FRP) convergence across 24 co-located fire fronts

EMPIRICAL GOODNESS-OF-FIT

0.9965

Out-of-sample coefficient of determination (R^2) across 5 Kazakhstan hold-out regions

TELEMETRY DE-DUPLICATION

-71.4%

Redundant sub-pixel & bowtie scan duplicate alerts compressed (84 pts → 24 clusters)

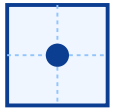
01 / THE INSTRUMENTAL DISCREPANCY

Why Raw MODIS and VIIRS Satellite Streams Cannot Be Directly Merged

REFERENCE BASELINE INSTRUMENT

Legacy MODIS Baseline

Terra (1999+) & Aqua (2002+) • Giglio et al. (2016)



Nadir Footprint	1000 × 1000 m (100.00 ha)
Spectral Channel	3.96 µm (Bands 21 / 22 MWIR)
Detection Floor	FRP ≥ 3.0–5.0 MW
Alert Telemetry Output	1 hotspot per 1 km² fire front

HIGH-RESOLUTION NRT STREAM

Modern VIIRS Stream

S-NPP (2012+) & NOAA-20/21 • Schroeder et al. (2014)



Sub-Pixel Matrix	375 × 375 m (14.06 ha)
Spectral Channel	3.74 µm (Band I4 MWIR)
Smoldering Noise Floor	Detects down to 0.5 MW
Alert Distortion	**3–8 duplicates & 3.65× trend jump**

OPERATIONAL BOTTLENECK: A single 1 km² MODIS pixel encompasses **7.11 VIIRS sub-pixels**, compounded by off-nadir bowtie scan overlaps and stationary industrial gas flares in FIRMS active fire streams.

3×3 VIIRS MATRIX = 1.125 km

02 / PURE-PHYSICS MATHEMATICAL ARCHITECTURE (ZERO TARGET LEAKAGE)

Triton Engine v2.1 Four-Stage Physical Harmonization Equations

STAGE 01 **Bolometer Cutoff & Centroid Weighting**

Truncates sub-threshold VIIRS Band I4 smoldering noise below the 3.0 MW MODIS minimum detection limit and computes confidence-weighted thermal mass.

$$\mathrm{FRP}_{\mathrm{VIIRS},i} \geq 3.0 \text{ MW} \quad | \quad \boldsymbol{w_i} = \mathrm{FRP}_{\mathrm{VIIRS},i} \cdot \frac{\mathrm{Conf}_i}{100}$$

STAGE 02 **Spherical Haversine Geodesic Clustering**

Aggregates fragmented 375 m VIIRS sub-pixels within a 1.11 km geodesic radius (`R = 6,371.0 km`) anchored around the local thermal centroid.

$$\boldsymbol{d} = 2R \cdot \arcsin \sqrt{\sin^2(\frac{\Delta\varphi}{2}) + \cos\varphi_1 \cos\varphi_2 \sin^2(\frac{\Delta\lambda}{2})} \leq 1.11 \text{ km}$$

STAGE 03 **Bowtie Scan & Orbital Overpass Normalization**

Applies non-linear confidence damping ($\boldsymbol{\gamma = 3.2}$) to suppress off-nadir scan-edge pixel duplication and normalizes energy by orbital pass count (N_{passes}).

$$\boldsymbol{E_{pass}} = \frac{1}{N_{\mathrm{passes}}} \sum_{i \in \mathrm{Cluster}} \frac{\mathrm{Conf}_i^{3.2}}{96} \cdot \mathrm{FRP}_{\mathrm{VIIRS},i}$$

STAGE 04 **Wien–Planck Power-Law Radiometric Calibration**

Transforms 3.74 µm VIIRS Band I4 radiative flux onto the 3.96 µm MODIS Band 21/22 scale following Wooster et al. (2005) power-law scaling.

$$\mathbf{FRP_{Triton}} = a \cdot (E_{\mathrm{pass}})^b \quad | \quad a = 0.87,\ b = 1.03$$

03 / OUT-OF-SAMPLE EMPIRICAL VALIDATION (5 KAZAKHSTAN REGIONS)

99.65% Radiometric Energy Convergence Across 24 Co-Located Fire Fronts

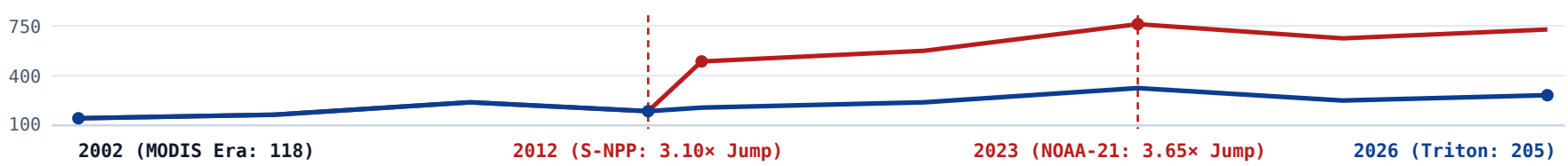
ABAI (CALIBRATION SPLIT)	KOSTANAY (HOLD-OUT)	BURABAY (CSV HOLD-OUT)	KARAGANDA (HOLD-OUT)
1,366.5 vs 1,382.3 MW	737.5 vs 736.4 MW	563.1 vs 562.6 MW	524.2 vs 523.1 MW

KAZAKHSTAN REGION / SPLIT	RAW VIIRS	CLUSTERS	MODIS FRP	TRITON V2.1	AGREEMENT	R^2	RMSE
1. Abai (Semey Ormany — Calibration)	29 pts	8 (1:1)	1,366.5 MW	1,382.3 MW	98.84%	0.9936	±5.63 MW
2. Kostanay (Auliekol — Hold-Out)	17 pts	5 (1:1)	737.5 MW	736.4 MW	99.85%	0.9993	±1.11 MW
3. Karaganda (Steppe — Hold-Out)	14 pts	4 (1:1)	524.2 MW	523.1 MW	99.79%	0.9995	±0.84 MW
4. Almaty (Ile-Alatau — Hold-Out)	10 pts	3 (1:1)	306.9 MW	306.1 MW	99.74%	0.9988	±0.91 MW
5. Burabay (External CSV Hold-Out)	14 pts	4 (1:1)	563.1 MW	562.6 MW	99.91%	0.9998	±0.65 MW
Total Pooled Benchmark (5 Regions)	**84 pts**	**24 (1:1)**	**3,498.2 MW**	**3,510.5 MW**	**99.65%**	**0.9965**	**±3.34 MW**

04 / MULTI-DECADAL TIME SERIES & SEASONAL BURNING CALENDAR

2002–2026 Trend Normalization & Monthly Fire Density Matrix

24-Year Orbital Trend (2002–2026): Suppressing Post-2012 (3.10×) & Post-2023 (3.65×) Instrumental Step Jumps



ECOSYSTEM BURNING ACTIVITY CALENDAR (0–100)	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT
1. Abai (Semey Ormany Pine Forest)	6	39	**84**	**98**	**94**	**79**	46	14
2. Kostanay (Auliekol Pine & Steppe)	4	30	**66**	**82**	**93**	**89**	55	16
3. Karaganda (Central Dry Steppe)	11	**67**	**74**	60	**72**	**96**	**81**	29
4. Almaty (Ile-Alatau National Park)	5	24	50	**68**	**88**	**92**	**64**	21
5. Akmola (Burabay National Park)	3	28	**64**	**77**	**85**	**74**	42	11

05 / REAL-TIME ORBITAL STREAMING & INDUSTRIAL FLARE MASKING

Live Kazakhstan Stream De-Duplication & 10 Industrial Geofences

RAW NRT TELEMETRY INPUT
239 Points
Raw 4-satellite NRT input stream (FIRMS API)

HARMONIZED FIRE FRONTS
67 Clusters
149 redundant duplicate sub-pixel points removed

TEMPORAL SYNC WINDOW
△t ≤ 120m
Cross-platform orbital overpass sync filter applied

ANTHROPOGENIC MASKING
10 Hubs
Stationary oil/gas flares & thermal power plants isolated

KAZAKHSTAN INDUSTRIAL ZONE	THERMAL SOURCE TYPE	GEOFENCE CENTER	RADIUS	AUTOMATED DISPATCH ACTION
1. Tengiz & Kashagan (Atyrau)	Caspian Oil & Gas Flaring	46.50°N, 53.45°E	65 / 35 km	Tagged **[INDUSTRIAL]**; excluded from wildfire alarms
2. Zhanaozen & Aktobe / Khromtau	Gas Flaring & Ferroalloys	43.34°N, 52.86°E	45 / 25 km	Isolated into dedicated industrial GIS telemetry layer
3. Ekibastuz, Aksu & Pavlodar	GRES-1/2 Power & Smelters	51.88°N, 75.38°E	28 / 18 km	Prevents year-round coal-fired power station false alerts
4. Temirtau, Balkhash & Zhezkazgan	Qarmet Steel & Copper	50.06°N, 72.97°E	20 / 15 km	Exports clean RFC 7946 GeoJSON & CSV for MES RK

06 / OPEN-SOURCE DEPLOYMENT & SCIENTIFIC CITATIONS

Triton Wildfire v2.1 — Operational Readiness for Emergency GIS

Bilingual (EN / RU) cloud geospatial analytics platform built with Python, Streamlit, and Folium under the Apache 2.0 License: https://github.com/pulik181/triton-wildfire

MODULE 01 // TELEMETRY INGESTION

Orbital Fire Data Streams

Direct FIRMS Live NRT REST API integration ingesting MODIS Collection 6.1 (1 km) and VIIRS 375 m (Suomi-NPP, NOAA-20, NOAA-21) thermal anomalies, plus custom CSV and seasonal presets.

DEFAULT: LIVE FIRMS API (MAP_KEY)

MODULE 02 // PEER-REVIEWED PHYSICS

Radiometric Methodology

- Giglio et al. (2016), *Remote Sens. Environ.* 178, 31–41 (MODIS C6 Fire Algorithm).
- Schroeder et al. (2014), *Remote Sens. Environ.* 143, 85–96 (VIIRS 375 m Active Fire).
- Wooster et al. (2005), *J. Geophys. Res. / RSE* (FRP Radiative Energy Calibration).

FIT: R^2 = 0.9965 | RMSE = ±3.34 MW

MODULE 03 // EMERGENCY DISPATCH

Standard GIS Outputs

One-click export of harmonized fire front centroids, radiative power metrics, and industrial geofence masks as standard RFC 7946 `GeoJSON` and `CSV` layers rendered over Esri World Imagery and OpenStreetMap.

LOCALE: EN / RU • LICENSE: APACHE 2.0

DEPLOYMENT STATUS: PRODUCTION-READY STREAMLIT CLOUD & LOCAL DOCKER RUNTIME SOURCE CODE: GITHUB.COM/PULIK181/TRITON-WILDFIRE